Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
		(vi)		based on graph in part (iv) total volume of solution heated = 25.0 + 28.0 = 53.0 (1) heat released = 53.0 × 4.18 × 12.0 = 2660 J (1) ecf possible when 25.0 or 28.0 used as the mass or when the maximum temperature is used instead of the temperature <u>increase</u>		2		2	1	
		(vii)		moles methanoic acid = $\frac{24.7}{46.02}$ = 0.537 (1) moles in 25.0 cm^3 = 0.0537 (1) enthalpy change = $-\frac{2660}{0.0537 \times 1000}$ = –49.5 kJ mol^{-1} (1) ecf possible when an attempt has been made to calculate a number of moles		3		3	2	
	(b)			more H^+ is available / there is a higher concentration of H^+ (1) award (1) for either of following hydrochloric acid is strong and methanoic acid is weak hydrochloric acid is stronger than methanoic acid neutral answers – more acidic / donates H^+ more easily	1		1	2		
	(c)	(i)		$CuCO_3(s) + 2HCOOH(aq) \rightarrow (HCOO)_2Cu(aq) + CO_2(g) + H_2O(l)$ formulae of reactants and products (1) state symbols and balancing (1)		2		2		

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